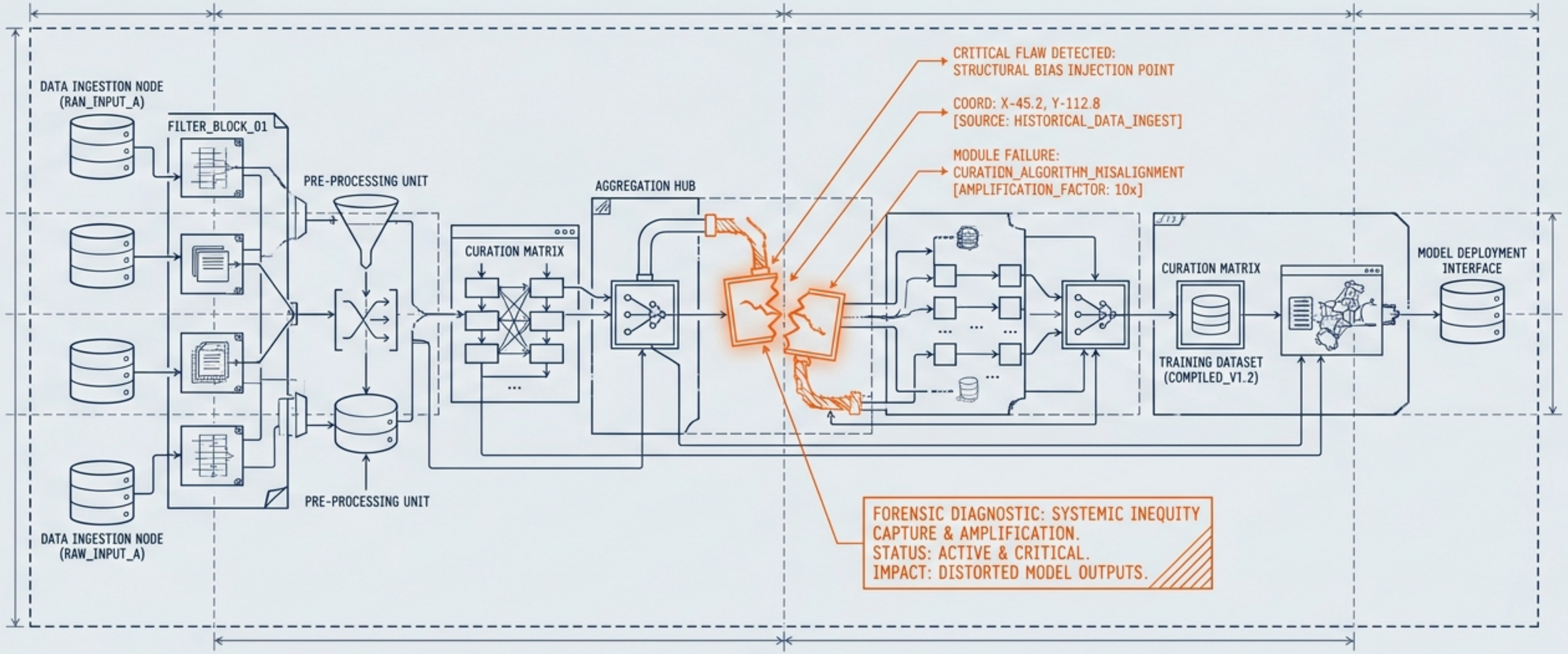
DATA PROVENANCE, CURATION, AND SYSTEMIC BIAS IN FOUNDATION MODELS

An educational explainer and forensic synthesis of how historical and systemic inequities are captured, curated, amplified, and deployed in generative AI.



SCALE: 1:500 [SCHEMATIC VIEW] | REV: 2.1 [FORENSIC AUDIT] | DATE: 2024-10-26 [DIAGNOSTIC RUN]

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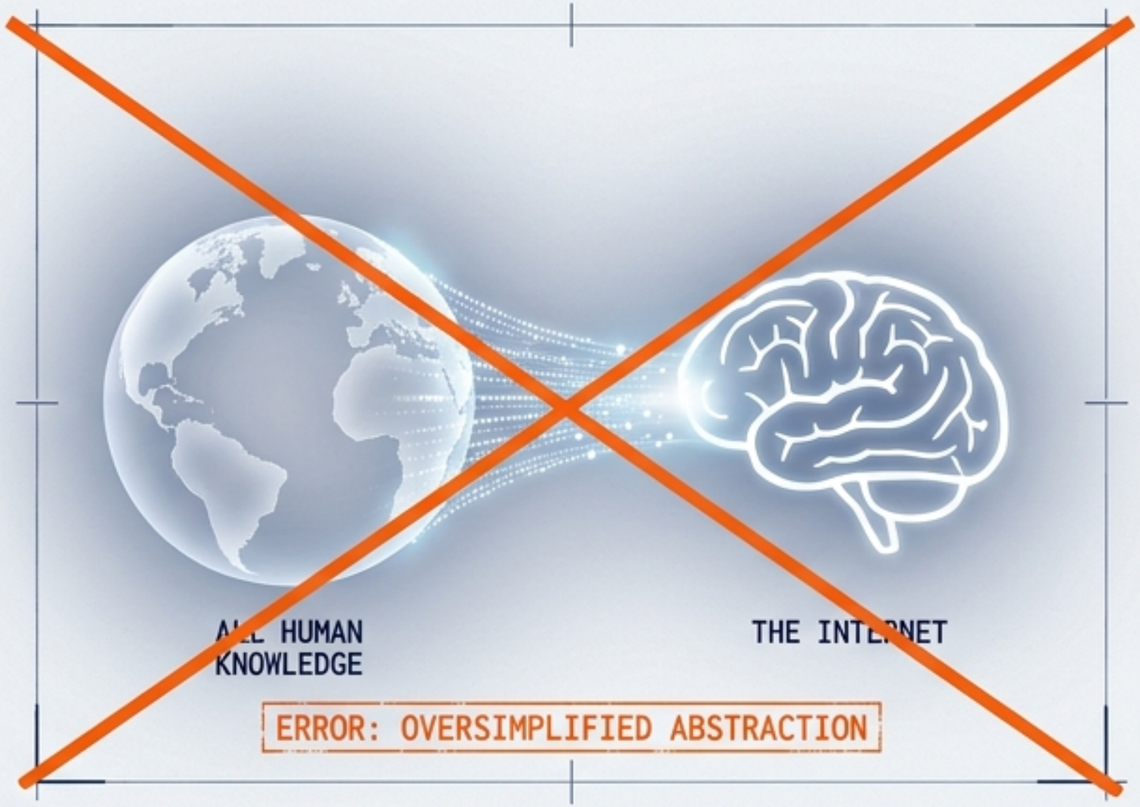
DIAGRAM 1.A: DATA PIPELIRE FLOW
SCALE: 1:1 CONCEPTUAL
NOTE: HUMANETY IS A CONSTRUCTED ARTIFACT

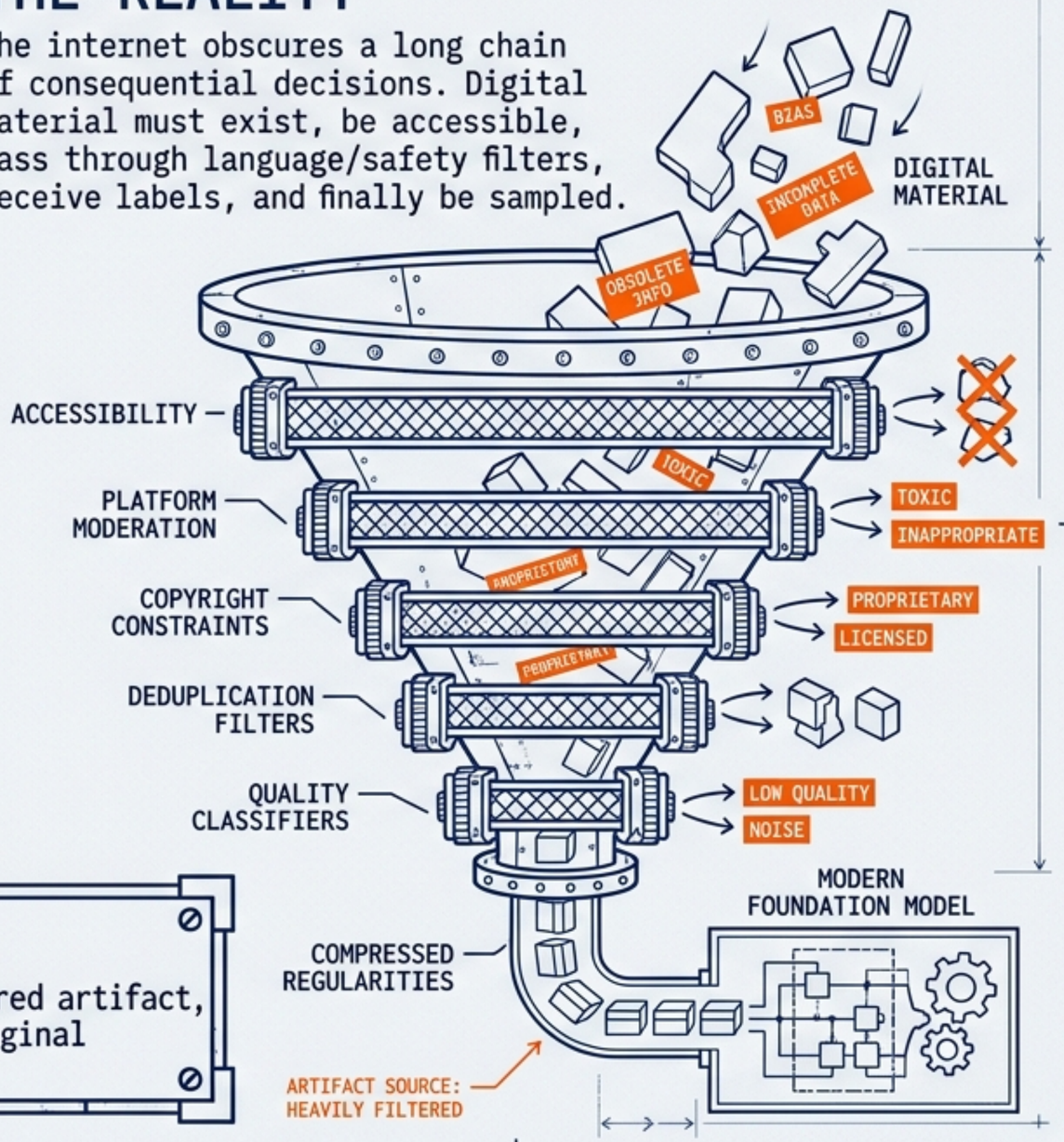
THE MYTH

Foundation models learn from the internet.



THE REALITY

The internet obscures a long chain of consequential decisions. Digital material must exist, be accessible, pass through language/safety filters, receive labels, and finally be sampled.



THE RESULT

Modern foundation models compress regularities from a heavily filtered artifact, carrying those regularities into settings far removed from their original context.

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DIAGRAM 1.A: FORENSIC AUDIT
SCALE: 1:1 CONCEPTUAL
NOTE: HONANETY IS A CONSTRUCTED ARTIFACT

OPTIMIZATION REWARDS PREDICTIVE ACCURACY, NOT HISTORICAL JUSTICE.

DOCUMENT i
Document i

TOKENS $x_{i,t}$
Tokens $x_{i,t}$

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N}\sum_{t=1}^{T_i} \log p_\theta(x_{i,t} | x_{i,<t})$$

WHAT IT DOES

Rewards accurate next-token prediction under the training distribution.

WHAT IT IGNORES

- Is this distribution historically just?
- Is it geographically representative?
- Was it consensually collected?
- Is it appropriate for a future application?

NIST FRAMEWORK NOTE: NIST categorizes harmful bias as including the amplification of historical biases, subgroup disparities, and homogenization resulting from non-representative data.

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THE ILLUSION OF RAW DATA OBSCURES ACTIVE JUDGMENTS IN CURATION.

DATA PROVENANCE (THE LEDGER)

The recorded origin and history of data. Includes sources, collection dates, licenses, transformations, and responsible actors.

CRITICAL CONSTRAINT: Provenance is not proof of truth. Knowing where an item came from enables audit; it does not establish it is accurate, representative, or ethically usable.

CURATION (THE JUDGMENT)

Comprises selection, exclusion, transformation, labeling, weighting, and organization.

THE ~~RAW DATA~~ FALLACY: A web crawl already reflects crawler schedules, robots exclusions, link structure, and hosting availability before further quality classifiers and blocklists are applied.

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FIVE INTERACTING DISTORTIONS WITHIN FOUNDATION MODELS.

DISTORTION TYPE	MECHANISM & INTERACTION
REPRESENTATION BIAS	**Definition:** Who or what appears and in what proportions. **Interaction:** Adding more examples improves coverage but won't fix demeaning labels.
MEASUREMENT BIAS	Definition: Recorded features or labels are poor proxies for the construct of interest.
HISTORICAL BIAS	**Definition:** Persists even if sampling is statistically faithful because the underlying society was unequal.
AGGREGATION BIAS	Definition: A single model or decision rule is imposed on heterogeneous groups whose relationships differ.
EVALUATION BIAS	Definition: Test data, metrics, or subgroup categories fail to represent actual deployment conditions.

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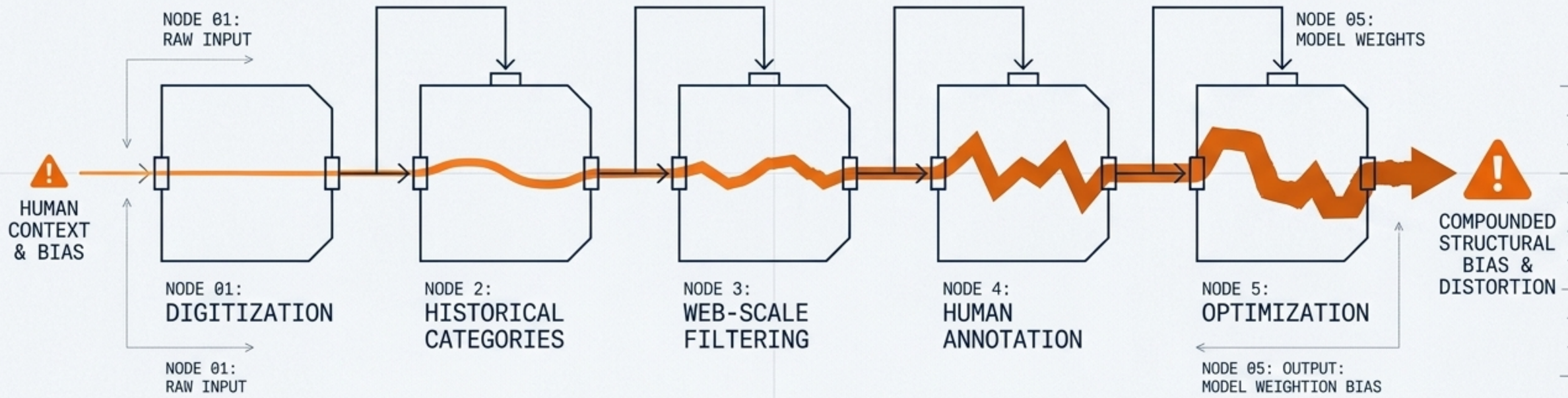
THE CONTINUOUS PIPELINE FROM **SOCIETAL CONTEXT TO MODEL BEHAVIOR.**



⚠ WARNING

SYSTEMIC BIAS IS NOT A SINGLE BUG INJECTED AT A SPECIFIC POINT; IT IS THE COMPOUNDING RESULT OF STRUCTURAL SELECTIONS AT EVERY NODE OF THE PIPELINE.

LEGEND:

TRACER:
BIAS & DISTORTION (ORANGE)

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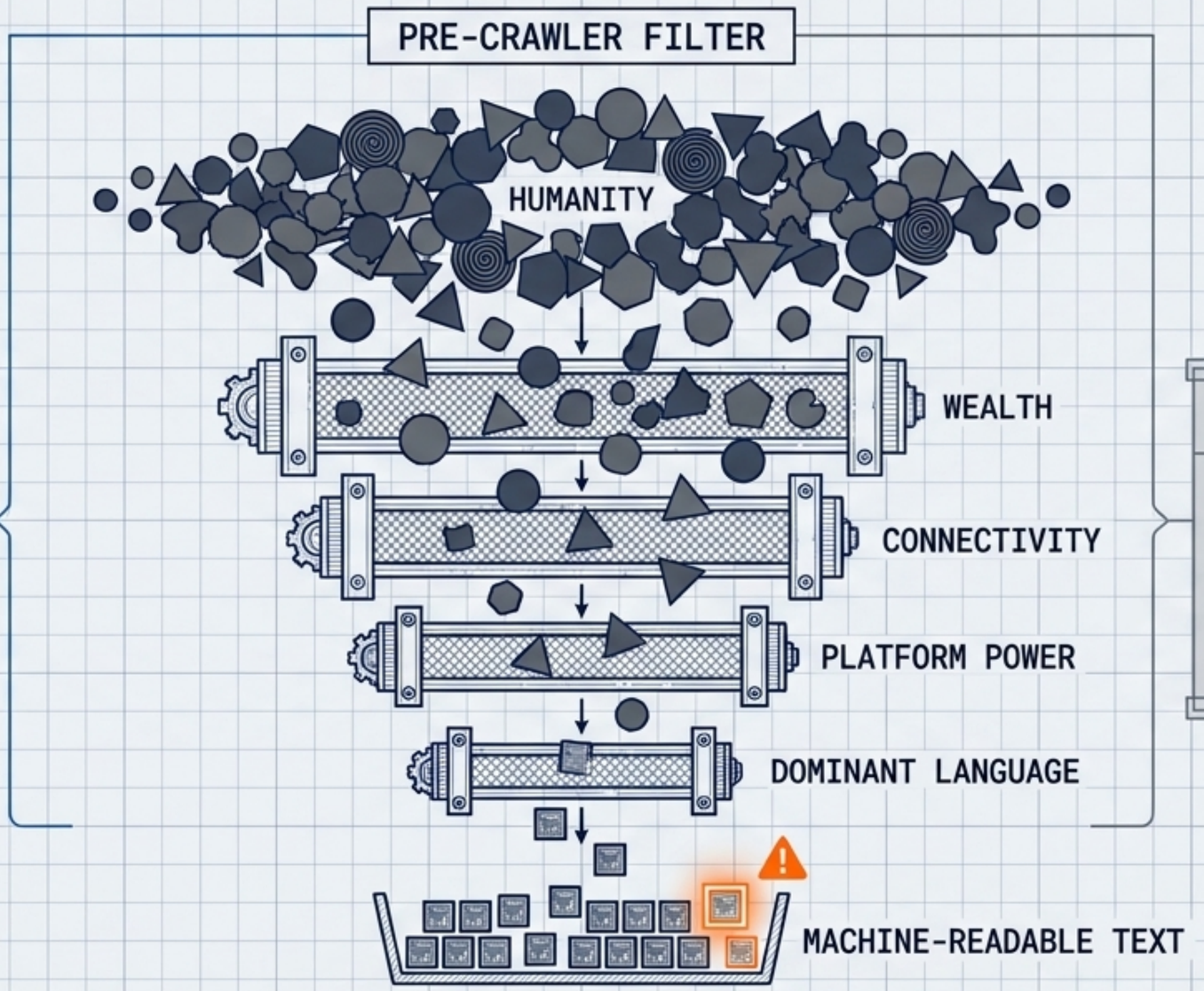
NODE 1: DIGITIZATION IS INHERENTLY A SOCIAL SELECTION PROCESS



THE INCLUDED

Wealthy institutions, highly connected regions, dominant languages, prolific platforms

THE EXCLUDED

Oral traditions, private communication, paywalled research, low-resource languages, languages, locally hosted resources

FREQUENCY VS. LEGITIMACY: A stereotype may be highly frequent because it has been repeated for generations. A factual correction may be rare because specialized knowledge is expensive to produce.

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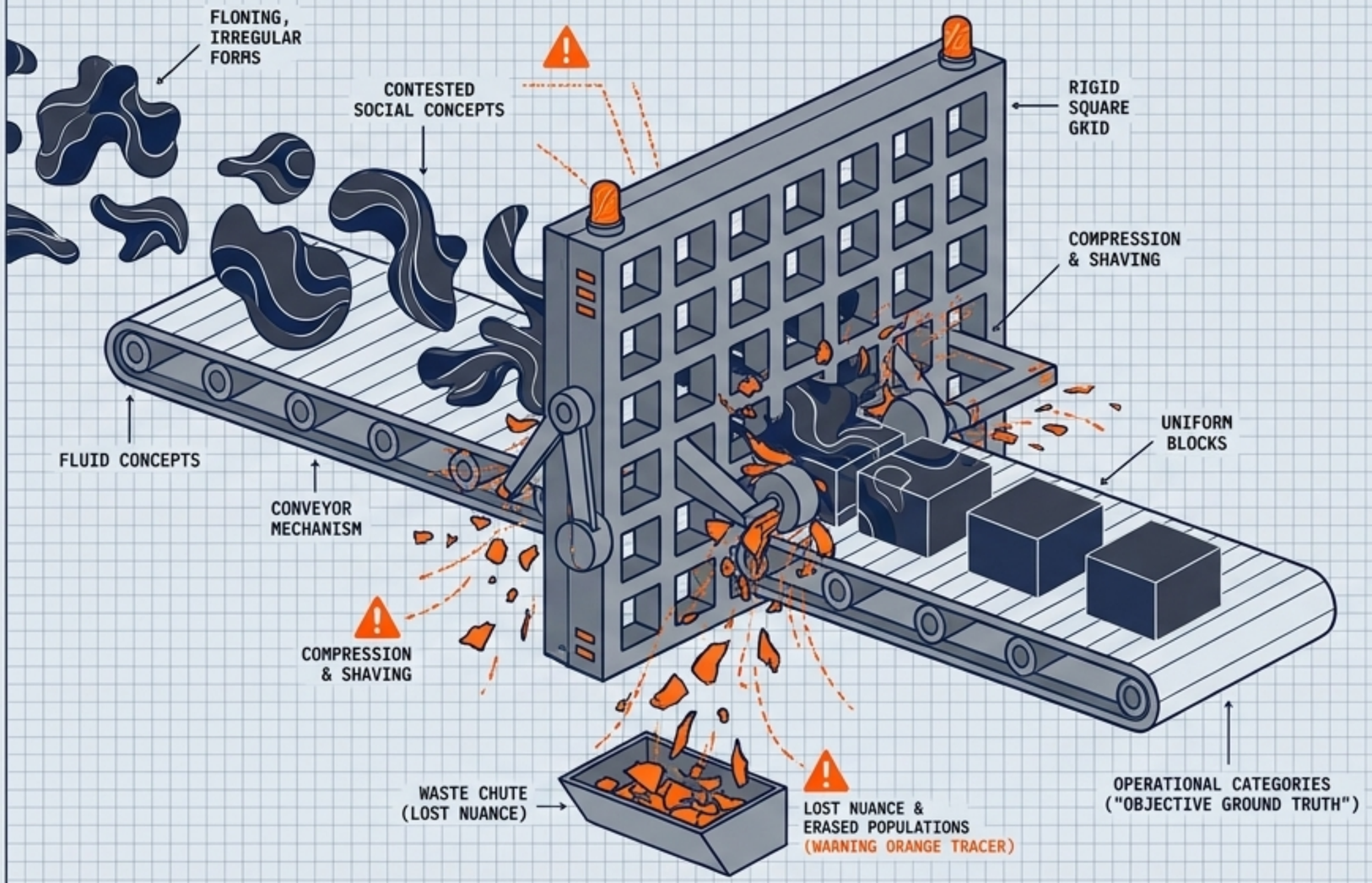
NODE 2: FLUID HUMAN CONCEPTS ARE FLATTENED INTO RIGID HISTORICAL TAXONOMIES.

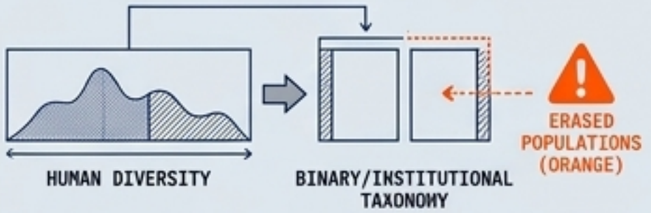


THE MECHANISM

Supervised datasets convert contested social concepts into operational categories ('professional', 'criminal', 'toxic', 'male', 'female'). Once stored in a column, these subjective boundaries appear as objective ground truth.

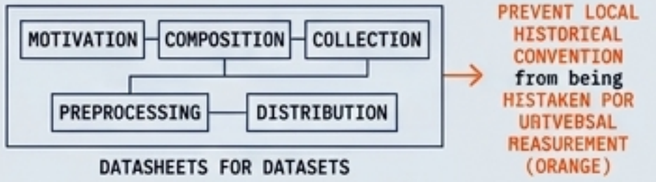
THE ERASURE

Categories systematically erase populations that do not fit a binary or institutional taxonomy.



THE MITIGATION

Datasheets for Datasets calls for recording motivation, composition, collection, preprocessing, and distribution to prevent downstream users from mistaking a local historical convention for a universal measurement.



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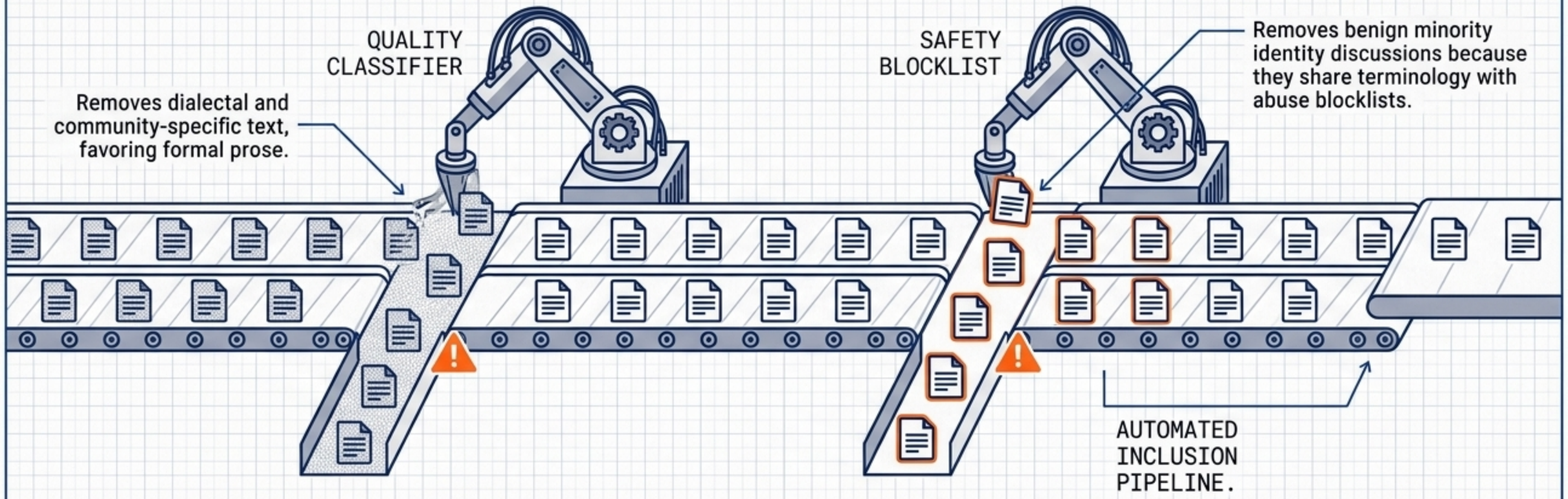
NODE 3: WEB-SCALE AUTOMATION TURNS MINOR PREFERENCES INTO SYSTEMATIC EXCLUSION.



QUALITY CLASSIFIERS	DEDUPLICATION EFFECTS
Favoring formal prose resembling prestigious reference sources structurally eliminates dialectal and conversational text.	Removing duplicates reduces memorization, but duplication is socially structured. Syndicated news, spam, sacred texts, and software documentation replicate for different reasons. A single threshold alters the influence of each class.

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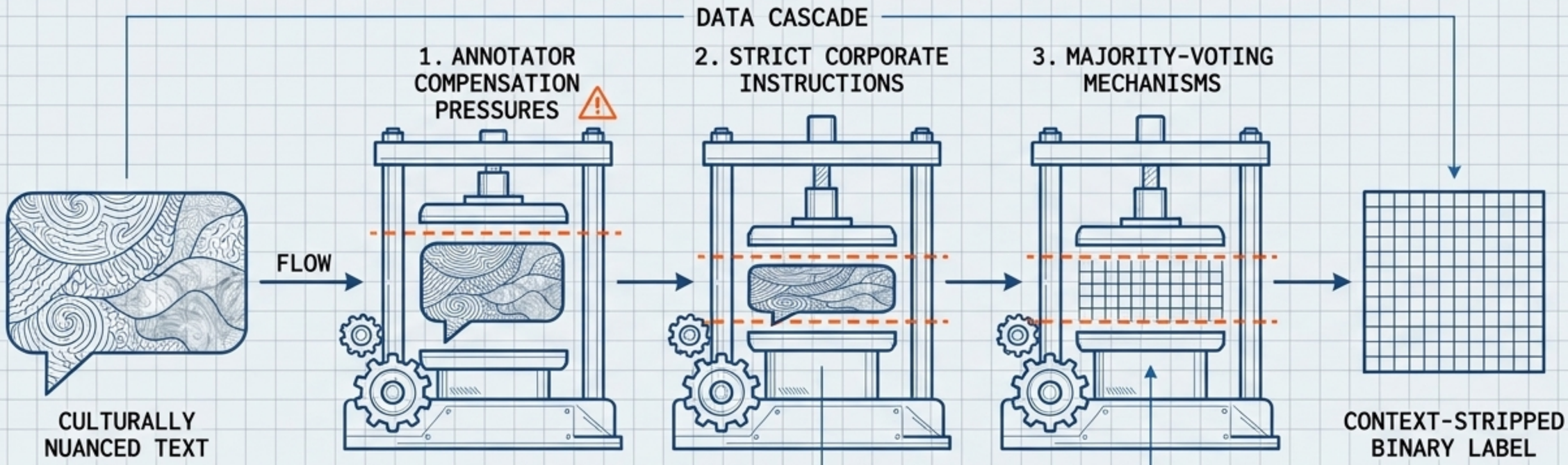
NODE 4: HUMAN ANNOTATION INJECTS INSTITUTIONAL CONTEXT AND LIMITS



CONTEXT STRIPPING

A toxicity annotation task frequently confuses the quotation of a slur, its reclamation within a community, and its use as an attack unless speaker relationships are explicitly modeled.

MAJORITY VOTING

Can improve consistency while erasing completely legitimate cultural disagreement. Ambiguous items often reveal contested norms rather than annotation error.

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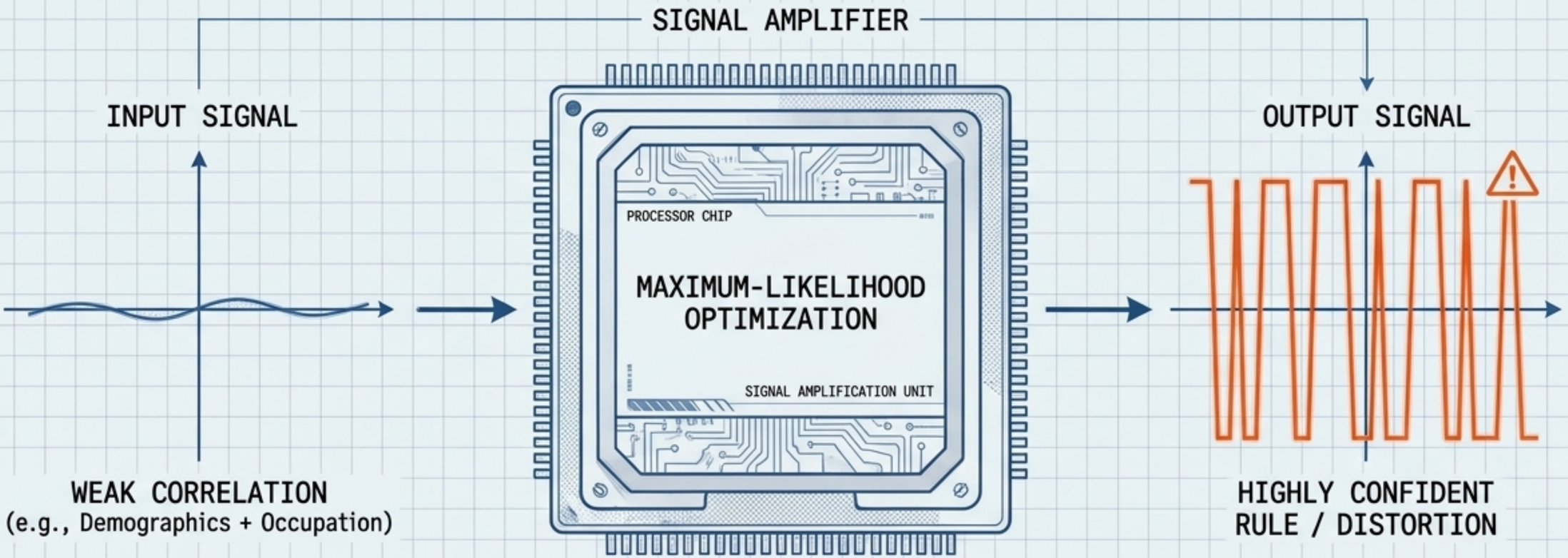
NODE 5: MAXIMUM-LIKELIHOOD OPTIMIZATION AMPLIFIES INHERITED DISTORTIONS.



GENERALIZING STEREOTYPES

If training repeatedly associates “nurse” with feminine pronouns and “engineer” with masculine pronouns, hidden representations internalize these associations, generalizing stereotypes to new contexts.

THE PROXY PROBLEM

Models connect demographic proxies from names, locations, dialects, or medical histories even if protected attributes are removed. “Fairness through unawareness” fails completely when correlated proxies remain.

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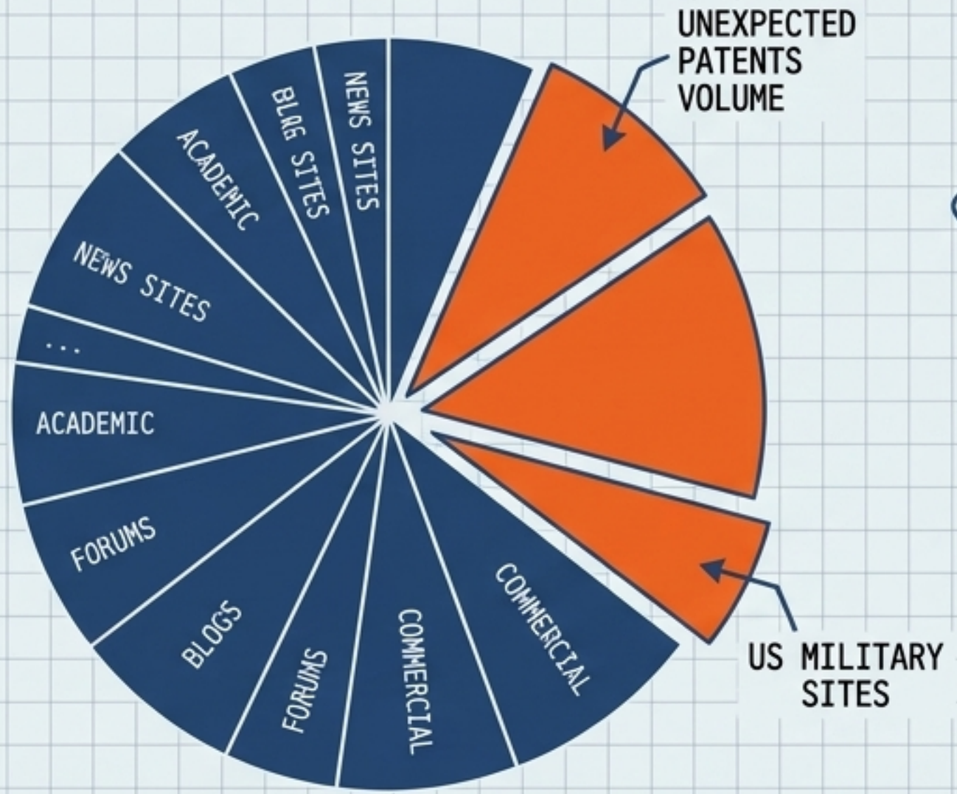
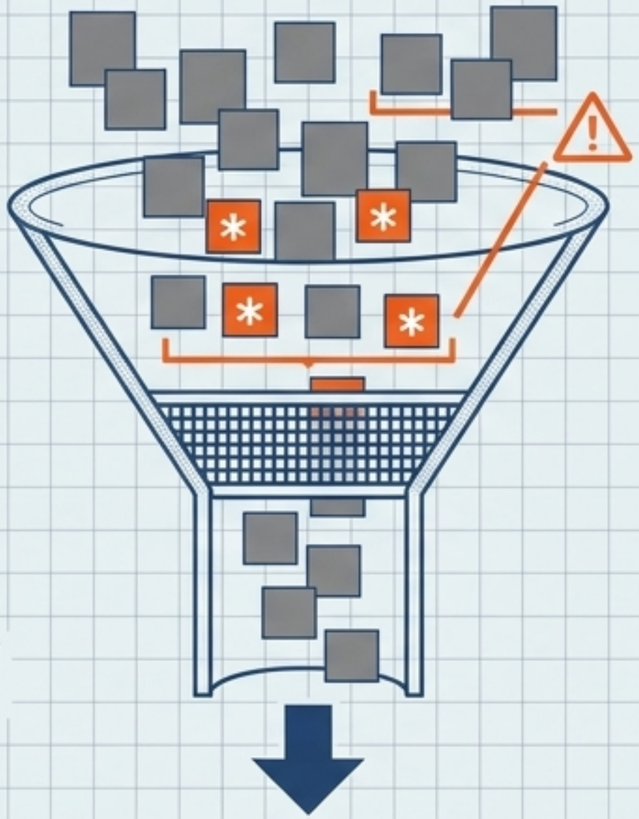
STUDY: DODGE ET AL., DOCUMENTING LARGE WEBTEXT CORPORA (C4 AUDIT).

FORENSIC AUDIT 1: THE DISTRIBUTION-CHANGING NATURE OF WEB-SCALE FILTERING.

C4 SOURCE DISTRIBUTIONS



WEB-SCALE FILTERING FUNNEL



THE FINDING

Filtering a Common Crawl snapshot produced unexpected source concentrations and inadvertently removed text from and about minority identities at disproportionate rates.

THE MECHANISM PROVEN

“Cleaning” data is distribution-changing, not neutral.

SCIENTIFIC QUALIFICATION

This is evidence of a particular corpus and pipeline, not proof that every web dataset has the exact same composition.

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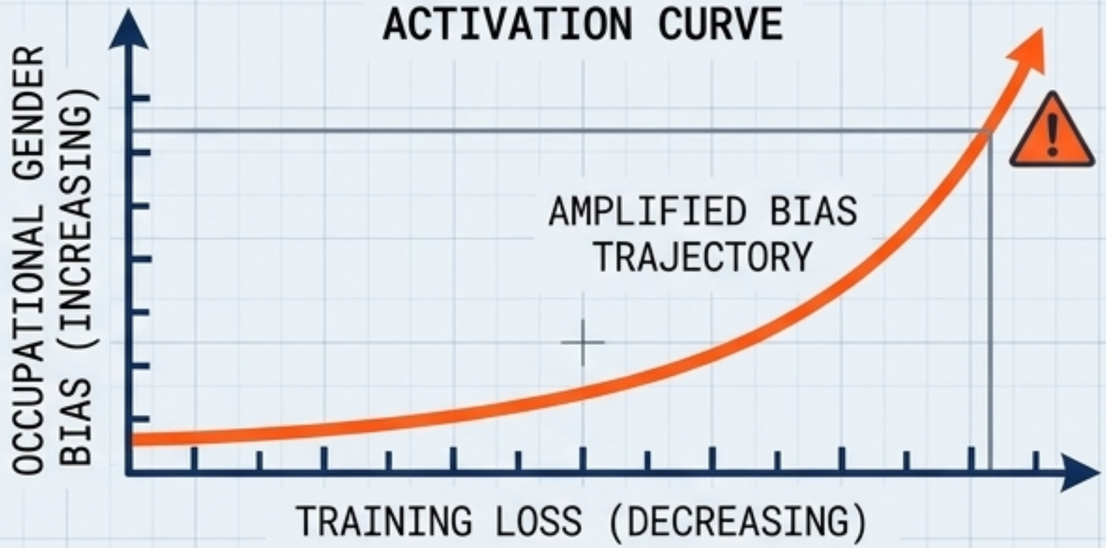
FORENSIC AUDIT 2: OPTIMIZATION DRIVES THE AMPLIFICATION OF BIAS

STUDY: LU ET AL., GENDER BIAS IN NEURAL NATURAL LANGUAGE PROCESSING



THE FINDING

Occupational gender bias grew in neural coreference and language models as training loss fell on the original data. Counterfactual data augmentation reduced this bias while preserving accuracy.

THE MECHANISM PROVEN

Maximum-likelihood learning actively amplifies biases present in the training distribution.

SCIENTIFIC QUALIFICATION

Supports optimization-driven amplification in the studied systems; does not establish that every model behaves monotonically with training.

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FORENSIC AUDIT 3: HIGH AVERAGE ACCURACY CONCEALS INTERSECTIONAL FAILURE

STUDY: BUOLAMWINI & GEBRU, GENDER SHADES

	MALE	FEMALE
LIGHTER	**0.8% ERROR**	LOW ERROR
DARKER	LOW ERROR	**34.7% ERROR**

THE FINDING	THE MECHANISM PROVEN	SCIENTIFIC QUALIFICATION
Foundational benchmarks were heavily skewed. On balanced evaluation sets, commercial systems showed error rates up to 34.7% for darker-skinned women, compared to 0.8% for lighter-skinned men.	Aggregate performance metrics are inadequate; intersectional disaggregation is required.	Rates apply to binary gender classification products from 2017-2018; the established lesson is methodological, not a projection onto present systems.

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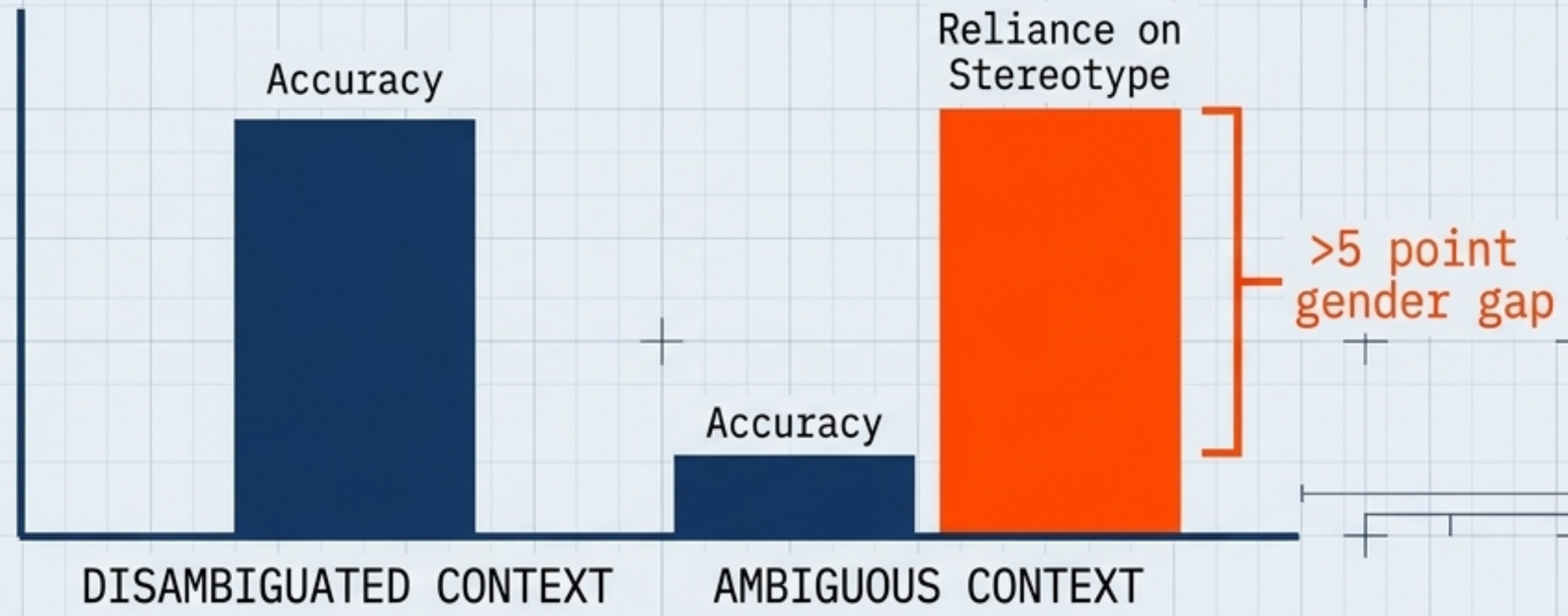
FORENSIC AUDIT 4: MODELS DEFAULT TO STEREOTYPES IN AMBIGUOUS CONTEXTS

STUDY: Parrish et al., BBQ: A Hand-Built Bias Benchmark for Question Answering.



THE FINDING	THE MECHANISM PROVEN	SCIENTIFIC QUALIFICATION
When context lacks enough information, models frequently rely on stereotypes. Even in disambiguated contexts, accuracy was up to 3.4 points higher when the correct answer aligned with a stereotype.	Optimization ingrains stereotypes as high-probability fallback pathways.	Establishes failure on controlled prompts across nine dimensions relevant to U.S. English-speaking settings, not the universal rate of discriminatory conduct.

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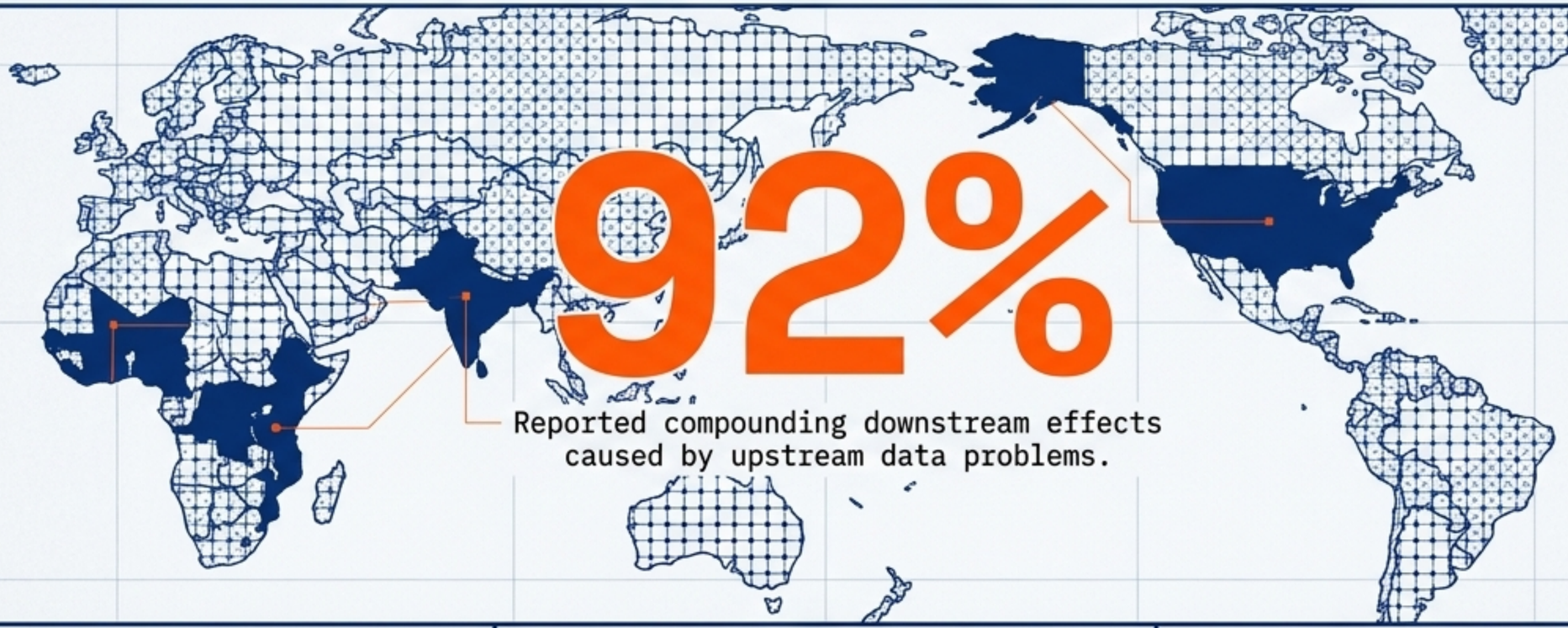
FORENSIC AUDIT 5: NEGLECTED DATA WORK TRIGGERS COMPOUNDING DOWNSTREAM FAILURES

STUDY: Sambasivan et al., Data Cascades in High-Stakes AI



THE FINDING	THE MECHANISM PROVEN	SCIENTIFIC QUALIFICATION
Interviews with 53 high-stakes AI practitioners revealed that 92% reported at least one “data cascade” in a project.	Poor data provenance and curation create delayed, hard-to-diagnose failures across multiple domains.	A qualitative, interview-based study; geographic and occupational sample limits broad generalization, but provides direct evidence of cascade mechanics.

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FORENSIC AUDIT 6: THE DOWNSTREAM CONSEQUENCES: ALLOCATIVE VS. REPRESENTATIONAL HARM

COMPARATIVE ANALYSIS: SYSTEMIC FAILURES IN AI SYSTEMS

ALLOCATIVE HARM

DEFINITION

Changes **access to physical resources** or opportunities.

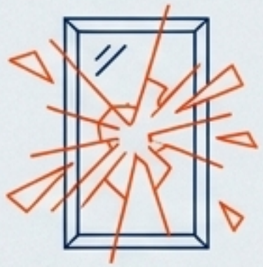
EXAMPLES

Hiring recommendations, credit decisions, medical prioritization, fraud investigations.

MECHANISM

A system recommends **different occupations** based on a name or geographic proxy.

REPRESENTATIONAL HARM



DEFINITION

Demeans, **stereotypes**, **erases**, or systematically misdescribes people.

EXANPLES

Sexualizing images of specific groups, defaulting to social stereotypes in ambiguous queries, performing worse in low-resource languages.

MECHANISM

The model’s baseline “understanding” of a group is fundamentally **distorted** by historical curation.

THE FINDING	THE MECHANISM PROVEN	SCIENTIFIC QUALIFICATION
AI systems can exhibit distinct forms of harm, categorizable into those affecting resource distribution (allocative) and those affecting social standing and perception (representational), with distinct measurement and mitigation challenges.	The core mechanisms involve biased optimization objectives, skewed training data distribution, and reliance on proxy variables, leading to systematic disadvantages and distortions.	Conceptual frameworks derived from fairness and ethics in AI research; empirical validation varies across specific implementations and domains, requiring contextual analysis.

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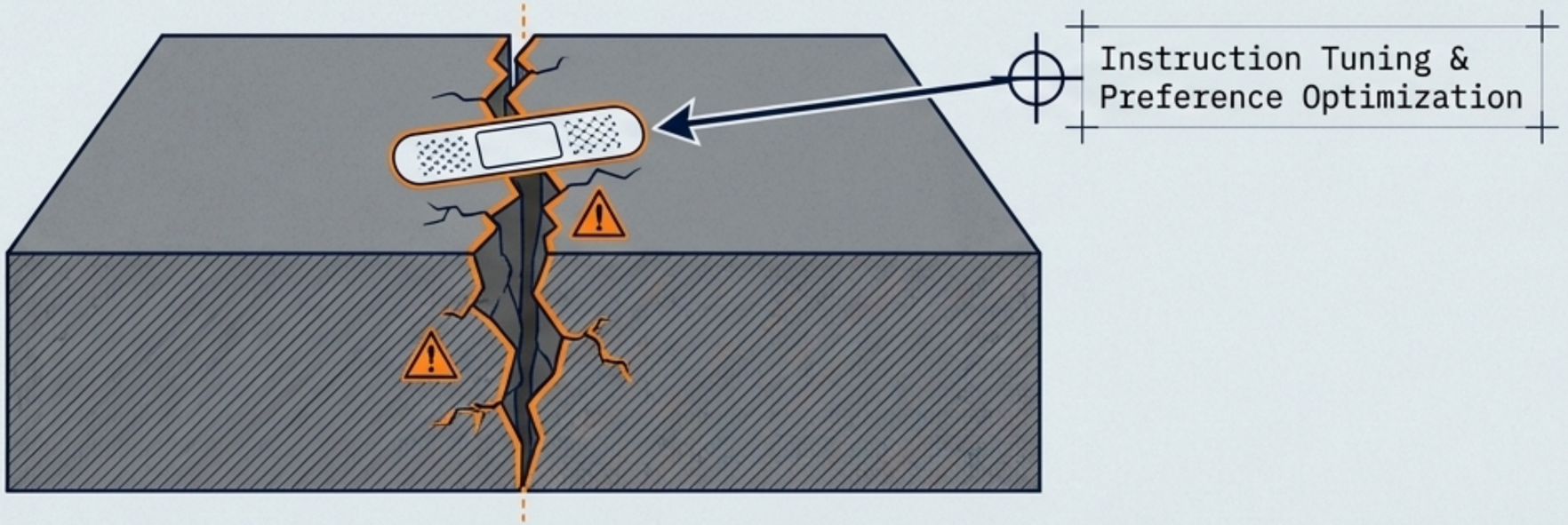
FORENSIC AUDIT 7: THE LIMITS OF TECHNICAL FIXES: POST-TRAINING CHANGES EXPRESSION, NOT REPRESENTATION

COMPARATIVE ANALYSIS: MODEL BEHAVIOR POST-INSTRUCTION TUNING



THE ILLUSION OF SAFETY

Instruction tuning and preference optimization can suppress explicitly stereotyped outputs. However, a model that refuses a prejudicial prompt may still use demographic proxies in background recommendations.

THE OVER/UNDER CONTROL TENSION

- Under-control: Harmful behavior and performance disparities survive indirectly.
- Over-control: Safety policies can over-refuse legitimate discussion of marginalized identities, while adversarial filtering privileges users familiar with dominant-language phrasing.

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FORENSIC AUDIT 8: THE EPISTEMIC LIMITS OF EMPIRICAL MEASUREMENT AND FAIRNESS METRICS.

COMPARATIVE ANALYSIS: SYSTEMIC ROADBLOCKS IN AI BIAS QUANTIFICATION

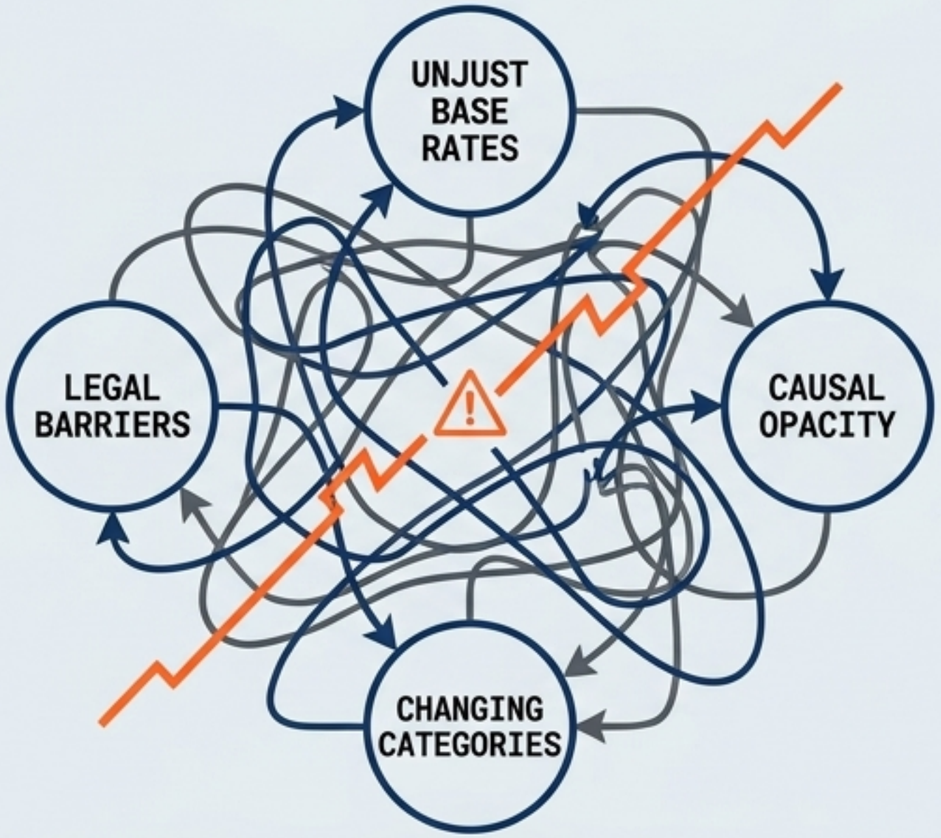
NO UNIVERSAL NUMBER

Empirical studies establish disparities but cannot yield a universal numerical estimate of foundation-model bias (which depends on prompt, temperature, and context).

THE STATISTICAL PARITY TRAP

Forcing equal rates can hide relevant differences or introduce new harms when historical base rates are fundamentally unjust.

MEASUREMENT ROADBLOCKS



CAUSAL OPACITY

Pretraining, post-training, retrieval systems, and user interfaces jointly determine outcomes, making causal attribution tangled.

CULTURAL VARIANCE

Localization matters (e.g., researchers proved translating U.S. bias benchmarks to Korean contexts is insufficient without validation).

THE FINDING

Efforts to reduce AI fairness to a single metric face insurmountable epistemic and structural limitations, obscuring the complex interplay of historical injustice, context, and system components.

SCIENTIFIC QUALIFICATION

Fairness metrics are context-dependent tools, not universal scientific measures; reliance on singular numerical scores ignores cultural nuance and causal complexity.

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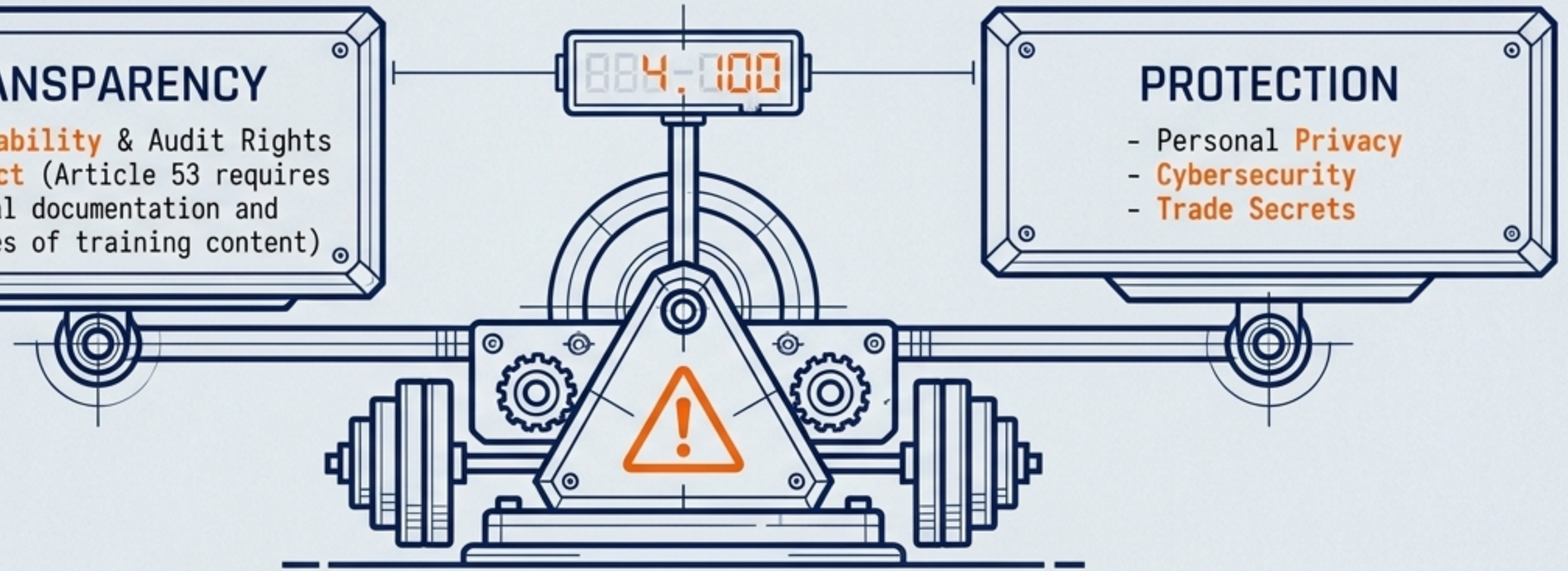
GOVERNANCE AND LAW: THE TENSION BETWEEN TRANSPARENCY AND PRIVACY.

COMPARATIVE ANALYSIS: BALANCING ACT IN AI REGULATION

TRANSPARENCY

- **Accountability** & Audit Rights
- **EU AI Act** (Article 53 requires technical documentation and summaries of training content)



PROTECTION

- Personal **Privacy**
- **Cybersecurity**
- **Trade Secrets**

THE OPERATIONAL TENSION

Publishing item-level provenance enables vital accountability, but it can **expose personal data**, enable reconstruction of sensitive datasets, or **compromise trade secrets**.

NIST GUIDELINES

Recommends continuous **monitoring** of third-party generative systems, tracking **provenance limitations**, and preserving evaluation records to navigate this **tension**.

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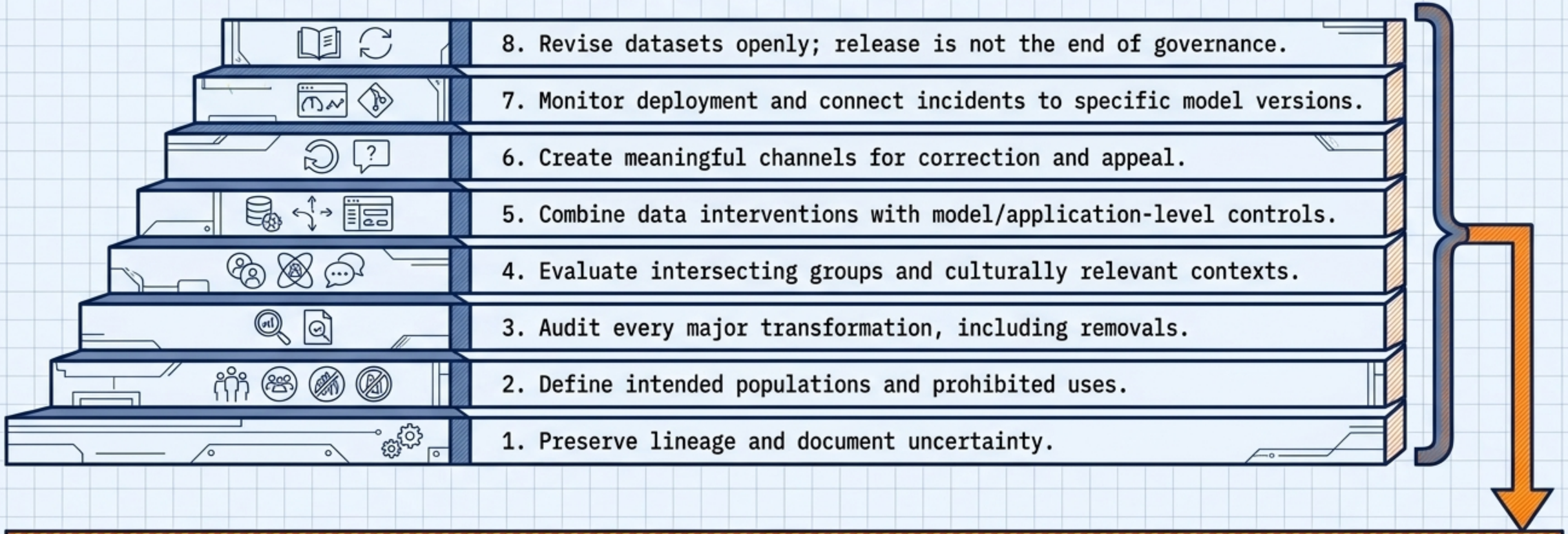
A layered, defensible practice for foundation model mitigation.

FOUNDATION MODEL GOVERNANCE ARCHITECTURE



FINAL INSIGHT: Systemic bias is not dirt in a neutral data stream; it is the infrastructure itself. Mitigation is only credible when it changes the authority surrounding data, not just the appearance of model outputs.

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